

Technical Documentation

VALVE.kicad_sch Control Board

Seeed Studio ESP32 113991054 Based PWM Servo and Valve Control PCB

Document Type	Engineering technical reference and wiring documentation
Source Schematic	VALVE.kicad_sch, KiCad E.D.A. 10.0.3, one-sheet schematic
Target MCU	Seeed Studio ESP32 113991054 / XIAO ESP32 architecture
Primary Function	PWM servo control, Pixhawk telemetry interface, I2C expansion, and servo power distribution
Power Domain	Battery input with onboard 5 V buck and selectable high-current external BEC rail
Revision	Documentation Rev. 1.0

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Contents

1	System Overview	2
2	System Architecture and Core Components	2
2.1	Microcontroller Unit	2
2.2	MCU Pin Utilization	3
3	Power Distribution Network	3
3.1	Battery Input and Fused Logic Supply	3
3.2	Onboard 5 V Buck Converter	4
3.3	Servo Power Selection	4
3.4	Recommended Servo Power Configuration	4
4	Signal Isolation and I2C Level Shifting	5
4.1	TXS0102DCT Level Translator	5
5	External Interfaces	5
6	Safety and Protection Mechanisms	6
6.1	Overcurrent Protection	6
6.2	Bulk Decoupling on Servo Rail	6
6.3	Signal Series Resistors	6
6.4	Status LED	6
7	Wiring and Harness Guidance	6
7.1	Pixhawk Telemetry Wiring	7
7.2	Servo Harness Orientation	7
7.3	External BEC Wiring Flow	7
8	Bring-Up and Test Procedure	7
8.1	Pre-Power Inspection	7
8.2	Logic Power Test	7
8.3	Servo Output Test	8
8.4	I2C Expansion Test	8
9	Electrical Verification Checklist	8
10	Schematic Reference	8
11	Firmware Interface Summary	10
12	Revision Notes	10

1 System Overview

The VALVE.kicad_sch control board is a custom power distribution and control PCB designed around the Seeed Studio ESP32 113991054 / XIAO ESP32 architecture. The board is intended to generate multiple servo PWM outputs, communicate with an external I2C servo expansion module, and interface with a Pixhawk flight controller telemetry port.

The architecture separates the sensitive logic supply from the high-current servo supply path. This allows the ESP32 and I2C logic to remain stable during high-load servo movement, while the servo rail can be powered either from the onboard 5 V buck converter or from an external high-current BEC/UBEC depending on the field configuration.

Design Note

The schematic uses two main power groups: a fused battery-fed logic buck path and a selectable servo power path. For field deployment, the external BEC option is recommended when the connected servos can draw stall currents higher than the onboard buck converter rating.

2 System Architecture and Core Components

2.1 Microcontroller Unit

Item	Schematic Reference	Description
MCU module	U1	Seeed Studio ESP32 113991054 module, used as the main controller for PWM, UART telemetry, I2C communication, manual switch input, and status indication.
Logic voltage	+3V3	ESP32 GPIO operates at 3.3 V logic level.
Module input supply	VUSB / +5V	The board routes the 5 V rail to the module input supply pin.
Unused pin	D9	Left unconnected to avoid accidental boot-strap or firmware conflict.

2.2 MCU Pin Utilization

Table 2: ESP32 pin allocation

MCU Pin	Net / Function	Protection / Interface	Notes
D0	PWM1	R1, 220 Ω series	Servo signal output for primary servo channel 1.
D1	PWM2	R2, 220 Ω series	Servo signal output for primary servo channel 2.
D2	PWM3	R3, 220 Ω series	Servo signal output for primary servo channel 3.
D3	PWM4	R4, 220 Ω series	Servo signal output for primary servo channel 4.
D4	SDA	TXS0102DCT level shifter	ESP32-side I2C data line at 3.3 V.
D5	SCL	TXS0102DCT level shifter	ESP32-side I2C clock line at 3.3 V.
D6	TX/RX interface	R5, 220 Ω series	Connected to Pixhawk telemetry header signal path. Verify TX/RX cross-over during harness assembly.
D7	RX/TX interface	R6, 220 Ω series	Connected to Pixhawk telemetry header signal path. Verify TX/RX cross-over during harness assembly.
D8	SW	J11 switch header	Manual override / discrete input. Intended to be pulled to GND when switch is closed.
D10	LED	R7, 330 Ω series	Onboard status LED output.
D9	NC	Not connected	Reserved / no connect.

Verification / Assembly Caution

The 220 Ω resistors on PWM and UART lines are signal-current limiting resistors. They help reduce pin stress and ringing, but they are not substitutes for flyback protection. Standard hobby servos contain their own motor electronics; the board only provides signal and power to the servo headers.

3 Power Distribution Network

3.1 Battery Input and Fused Logic Supply

The main battery input enters the board through connector J3, an XT60-M connector. The schematic labels the expected input range as approximately 8 V to 28 V. This rail is named VBAT_IN. The logic path passes through fuse F1 before feeding the onboard buck converter input rail VBAT_FUSED.

Table 3: Main power path

Stage	Reference / Net	Function
Battery connector	J3 / VBAT_IN	XT60-M raw battery input, marked 8 V to 28 V.
Fuse	F1 / VBAT_FUSED	5 A surface-mount fuse protecting the logic buck input path.
Buck converter	U3 / TPS54302	Synchronous buck converter generating the regulated +5V rail.
Output inductor	L1 / 10 μ H	Energy storage inductor for buck conversion.
Bulk and ceramic capacitors	C1, C2, C6, C8, C9, C10	Input, bootstrap, output, and compensation/feedback support components around the buck regulator.

3.2 Onboard 5 V Buck Converter

The U3 TPS54302 regulator produces the +5V system rail. This rail powers the ESP32 module input supply, the high-side voltage of the I2C level shifter, and the external I2C expansion connector.

Design Note

Although the TPS54302 can supply a multi-ampere 5 V rail, the onboard buck should be treated as the logic and light auxiliary rail unless thermal and current testing confirms that it can safely power the selected servos in the final enclosure.

3.3 Servo Power Selection

The board supports a selectable servo rail called +VCC. This rail feeds the four 3-pin servo headers. Connector J7 is used as a power selection point between the onboard +5V rail and the external BEC rail EXT_BEC.

Table 4: Servo power selection network

Reference	Net / Connector	Description
JP1	VBAT_IN to external BEC input path	Open jumper pad for feeding raw battery voltage to the external BEC input connector. Populate or bridge only when the external BEC wiring scheme requires board-side battery feed.
J5	External BEC IN	Two-pin connector carrying battery input and GND to an external BEC/UBEC module.
J8	External BEC OUT	Two-pin connector accepting the regulated external BEC output back into the board as EXT_BEC.
J7	Power selection port	Three-position selection header allowing +VCC to be linked to either onboard +5V or EXT_BEC.
+VCC	Servo supply rail	High-current supply net distributed to the servo headers J4, J6, J9, and J10.

3.4 Recommended Servo Power Configuration

- Use the onboard +5V rail for bench testing, low-current logic loads, or very small servos only.
- Use EXT_BEC for flight or payload operation where servo stall current can exceed the onboard regulator thermal capability.

- Keep all grounds common: Pixhawk GND, ESP32 GND, external BEC GND, and servo GND must share the same reference on the board.
- Size the external BEC according to total worst-case stall current, not only normal running current.

4 Signal Isolation and I2C Level Shifting

4.1 TXS0102DCT Level Translator

Component U2 is a TXS0102DCT bidirectional voltage-level translator. It shifts ESP32-side 3.3 V I2C signals to 5 V bus signals for external expansion.

Table 5: I2C level shifting architecture

Signal	Voltage Domain	Routing
SDA / SCL	3.3 V	ESP32 pins D4 and D5 connect to U2 side A.
SDA1 / SCL1	5 V	U2 side B connects to the PCA9685 / I2C expansion header J2.
VCCA	3.3 V	Low-voltage reference for ESP32-side logic.
VCCB	5 V	High-voltage reference for external I2C logic.
OE	Logic enable	Pulled to the appropriate enable domain according to the schematic.
R10, R11	4.7 k Ω	Pull-up resistors on SDA1 and SCL1.

Verification / Assembly Caution

For longer I2C cable runs, keep the bus speed conservative, typically 100 kHz, and verify signal edges with an oscilloscope. Very long external cables may require a more robust differential I2C extender instead of a simple level shifter.

5 External Interfaces

Table 6: Connector and interface map

Designator	Connector / Type	Function	Pinout / Notes
J1	4-pin header	Pixhawk telemetry	TX, RX, GND reference connections. Cross-check Pixhawk TELEM pinout so board TX goes to Pixhawk RX and board RX goes to Pixhawk TX.
J2	PCA9685 / I2C expansion header	External I2C servo expansion	Provides SDA1, SCL1, +5V, and GND. The schematic symbol is a multi-pin header; unused pins should be documented on the harness drawing.
J3	XT60-M	Main battery input	Raw input VBAT_IN, marked 8 V to 28 V.
J4	3-pin servo header	Servo channel 3	PWM3, +VCC, GND.
J6	3-pin servo header	Servo channel 2	PWM2, +VCC, GND.
J9	3-pin servo header	Servo channel 1	PWM1, +VCC, GND.

Designator	Connector / Type	Function	Pinout / Notes
J10	3-pin servo header	Servo channel 4	PWM4, +VCC, GND.
J5	2-pin socket	External BEC input	Battery input feed and GND for optional external BEC.
J8	2-pin socket	External BEC output	Regulated BEC output returned to the board as EXT_BEC.
J7	3-pin socket	Servo power selection	Jumper/bridge selection for +VCC source: onboard +5V or EXT_BEC.
J11	2-pin header	Switch override input	Pulls D8/SW to GND when closed. Add firmware pull-up or external pull-up as required.
H1–H4	Mounting holes	Mechanical mounting	Connected to GND in the schematic.

6 Safety and Protection Mechanisms

6.1 Overcurrent Protection

Fuse F1 is specified as 0451005.MRL, rated 5 A. It protects the onboard logic regulator input path from catastrophic short-circuit conditions on the fused rail. It does not protect the full raw battery path if JP1 or external BEC wiring bypasses the fused section.

6.2 Bulk Decoupling on Servo Rail

Capacitors C5 and C7 are 1000 μ F electrolytic capacitors placed on the servo supply rail +VCC. These capacitors reduce local voltage droop during servo startup, reversal, and stall events.

6.3 Signal Series Resistors

Resistors R1 through R6 are 220 Ω series components on direct MCU signal lines. Their role is to limit transient current, reduce ringing, and provide a small amount of fault isolation between the external harness and ESP32 GPIO pins.

6.4 Status LED

D1, driven by D10 through R7 (330 Ω), provides onboard firmware status indication. Recommended LED states are listed in Table 7.

Table 7: Recommended firmware LED states

LED Behaviour	Meaning
Fast blink at boot	Firmware alive, board initializing.
Slow blink	Normal idle state, Pixhawk link not required or not yet detected.
Solid ON	Pixhawk telemetry and I2C expansion detected.
Three quick pulses	Manual switch override event detected on D8.
Rapid continuous blink	Fault condition: low voltage, I2C failure, or servo rail not armed.

7 Wiring and Harness Guidance

7.1 Pixhawk Telemetry Wiring

Table 8: Pixhawk TELEM harness rule

Board Side	Connect To Pixhawk Side
Board TX	Pixhawk RX
Board RX	Pixhawk TX
Board GND	Pixhawk GND
+5V	Do not connect unless the Pixhawk port is intentionally powering or being powered by this board. Avoid back-feeding.

7.2 Servo Harness Orientation

Each servo header carries signal, power, and ground. Before first power-up, verify that the physical connector orientation matches the silkscreen. Reversing +VCC and GND can permanently damage servos and the board.

7.3 External BEC Wiring Flow

1. Connect battery input to J3.
2. If the external BEC is to be fed through the board, bridge JP1 only after verifying the trace current rating and BEC input current requirement.
3. Connect J5 to the input of the external BEC.
4. Connect the regulated BEC output back to J8.
5. Fit the J7 selection jumper so +VCC is sourced from EXT_BEC.
6. Confirm +VCC voltage with a multimeter before installing any servo.

8 Bring-Up and Test Procedure

8.1 Pre-Power Inspection

1. Check all solder joints around U3, U2, J3, F1, J7, J8, and servo headers.
2. Confirm there is no short between VBAT_IN and GND.
3. Confirm there is no short between +5V and GND.
4. Confirm the power selection header J7 is not accidentally shorting both +5V and EXT_BEC together.
5. Set the bench power supply current limit before applying battery voltage.

8.2 Logic Power Test

1. Apply 8 V to 12 V from a current-limited bench supply to J3.
2. Measure VBAT_IN; it should match the supply input.
3. Measure VBAT_FUSED; it should be present after F1.
4. Measure +5V; it should be close to 5.0 V.
5. Measure +3V3 on the ESP32 module; it should be close to 3.3 V.
6. Program a basic LED blink firmware and verify D1 on D10.

8.3 Servo Output Test

1. Test PWM outputs using an oscilloscope or logic analyzer before connecting a servo.
2. Confirm a 50 Hz servo PWM frame or the required servo refresh rate.
3. Confirm pulse width range in firmware, for example 1000 μ s to 2000 μ s for standard RC servos.
4. Connect one servo at a time with a current-limited supply.
5. Increase load gradually and monitor +VCC droop and board temperature.

8.4 I2C Expansion Test

1. Connect the external PCA9685 or I2C expansion board to J2.
2. Run an I2C scanner at 100 kHz.
3. Confirm the expected device address is detected.
4. If the bus fails, check SDA1, SCL1, pull-ups R10/R11, and common GND.

9 Electrical Verification Checklist

Table 9: Board-level verification checklist

Test Item	Expected Result	Action if Failed
Input rail continuity	J3 positive reaches VBAT_IN	Inspect XT60 soldering and input trace.
Fuse path	VBAT_FUSED present after F1	Check fuse orientation, rating, and soldering.
Buck output	+5V near 5.0 V	Inspect U3, L1, feedback divider, output capacitors, and enable network.
ESP32 power	Module boots and +3V3 is present	Verify +5V input, USB/VUSB routing, module orientation, and solder joints.
PWM pins	Clean pulse train on PWM1–PWM4	Check firmware pin mapping and R1–R4 continuity.
UART wiring	Board TX/RX communicate with Pixhawk	Swap TX/RX if no serial data is received. Confirm common GND.
I2C level shift	Device detected on SDA1/SCL1	Check U2 supplies, OE pin, pull-ups, cable length, and bus speed.
Switch input	D8 changes state when J11 is closed	Enable internal pull-up in firmware or add external pull-up if needed.
Servo rail selection	+VCC equals selected source	Verify J7 jumper position and confirm that both sources are not tied together.
Servo decoupling	No large brownout under servo motion	Use external BEC, increase bulk capacitance, shorten high-current wiring, or reduce servo load.

10 Schematic Reference

Figure 1 shows the provided one-sheet schematic reference used for this documentation.

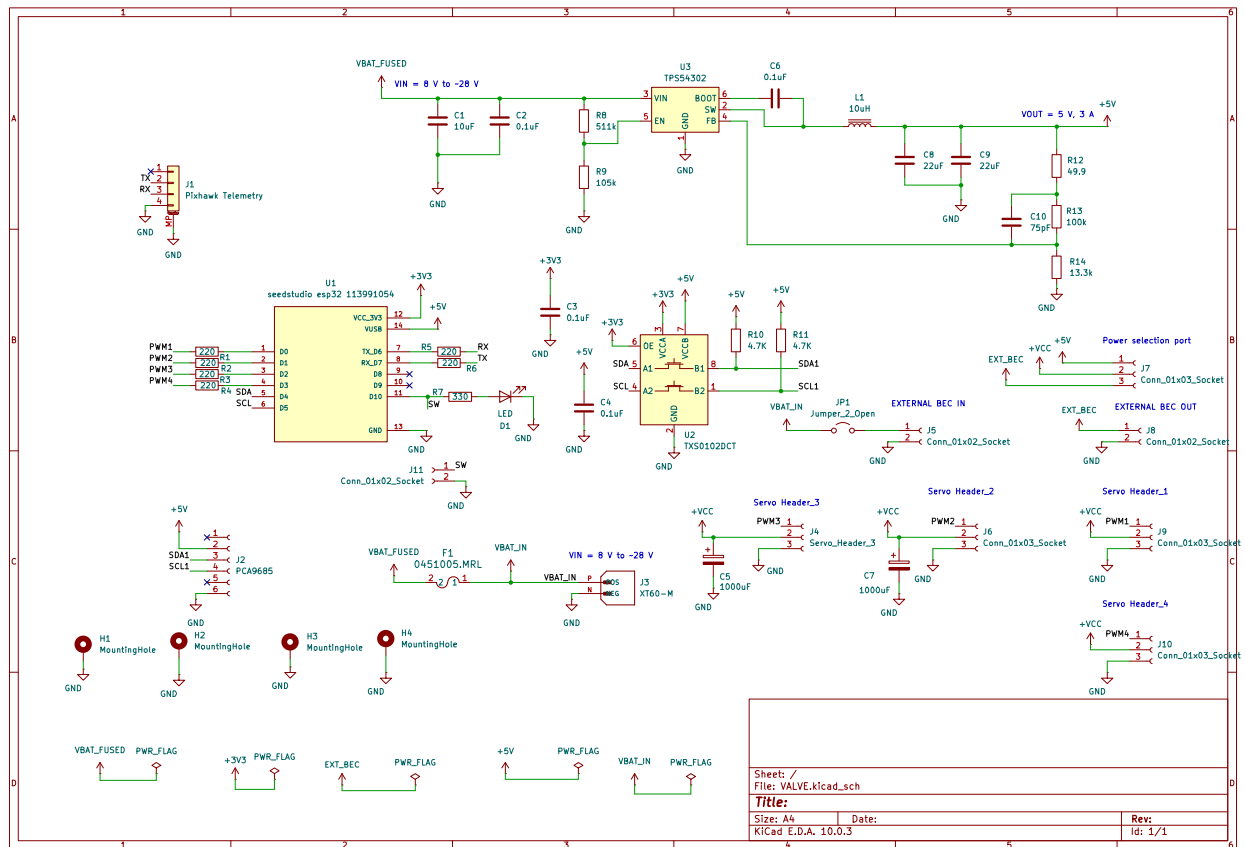


Figure 1: schematic

11 Firmware Interface Summary

Table 10: Recommended firmware abstraction

Firmware Object	Hardware Pins	Recommended Function
ServoChannel[1]	D0 / PWM1	Main valve or servo channel 1.
ServoChannel[2]	D1 / PWM2	Main valve or servo channel 2.
ServoChannel[3]	D2 / PWM3	Main valve or servo channel 3.
ServoChannel[4]	D3 / PWM4	Main valve or servo channel 4.
I2CExpansion	D4 SDA, D5 SCL	PCA9685 expansion bus.
PixhawkSerial	D6, D7	MAVLink / simple telemetry / command interface.
ManualOverride	D8	Active-low manual switch input.
StatusLED	D10	Board status and fault indication.

12 Revision Notes

Table 11: Documentation revision history

Revision	Date	Change Summary
1.0	June 10, 2026	Initial technical documentation generated from provided VALVE.kicad_sch schematic and design notes.